

The Variogram: Quantifying Spatial Dependence

Spatial Data Science - Graduate Lecture

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Lecture Outline (90 minutes)

- 1 Motivation & Intuition (15 min)
- 2 Foundations: Randomness & Stochastic Processes (20 min)
- 3 Theoretical Foundation (25 min)
- 4 Empirical Variogram Estimation (20 min)
- 5 Variogram Modeling (20 min)
- 6 Anisotropy & Demonstration (15 min)
- 7 Common Mistakes & Pitfalls (10 min)
- 8 Discussion & Q&A (5 min)

Why Do We Need the Variogram?

Tobler's First Law of Geography

"Everything is related to everything else, but near things are more related than distant things."

Limitations of covariance/correlation:

- Require second-order stationarity (constant mean, constant variance, covariance depends only on lag)
- Often violated in real spatial data
- Covariance may not exist for intrinsic processes

Key idea: The variogram works under weaker assumptions.

Intuitive Example: Soil pH

Two nearby points (distance = 5 m)

- pH = 6.2, 6.3
- Difference = 0.1

Two far apart points (distance = 200 m)

- pH = 6.2, 7.1
- Difference = 0.9

Dissimilarity
increases with
distance

Goal: Find a function that quantifies this increase in dissimilarity.

What Is a Random Spatial Process?

Deterministic vs Random

- **Deterministic:** Given input, output is perfectly predictable. Example: $y = 2x + 1$
- **Random (Stochastic):** Even with same input, outcome varies unpredictably.

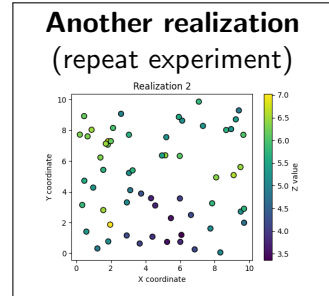
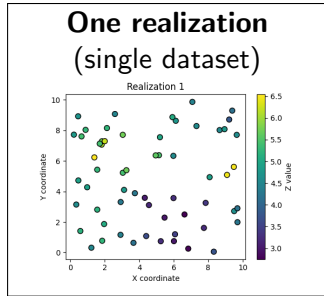
Spatial context:

$$\underbrace{Z(\mathbf{s})}_{\text{Value at location } \mathbf{s}} = \underbrace{m(\mathbf{s})}_{\text{Deterministic trend}} + \underbrace{\varepsilon(\mathbf{s})}_{\text{Random component}}$$

Example: Soil zinc concentration $Z(\mathbf{s})$ at location $\mathbf{s}$:

- $m(\mathbf{s})$: large-scale trend (e.g., decreases with elevation)
- $\varepsilon(\mathbf{s})$: small-scale random variation + measurement error

Realizations of a Spatial Process



- Each dataset is one **realization** of the same underlying stochastic process
- If we could re-sample the entire study area, we would get a different realization
- **Problem:** In spatial statistics, we typically see only **one realization**
- **Solution:** Make assumptions about **stationarity** — certain properties are constant across space

Key Concepts: Random Function and Its Moments

Definition

A **random function** $Z(\mathbf{s})$ assigns a random variable to every location $\mathbf{s}$ in the domain.

First-order moment (mean function):

$$\mu(\mathbf{s}) = \mathbb{E}[Z(\mathbf{s})]$$

The expected value at location $\mathbf{s}$ (average over all realizations).

Second-order moment (covariance function):

$$C(\mathbf{s}_1, \mathbf{s}_2) = \text{Cov}(Z(\mathbf{s}_1), Z(\mathbf{s}_2)) = \mathbb{E}[(Z(\mathbf{s}_1) - \mu(\mathbf{s}_1))(Z(\mathbf{s}_2) - \mu(\mathbf{s}_2))]$$

Problem: With only one realization, we cannot estimate $\mu(\mathbf{s})$ or $C(\mathbf{s}_1, \mathbf{s}_2)$ directly — we need stationarity.

Why Stationarity? The Replication Problem

In classical statistics:

Observe $X_1, X_2, \dots, X_n$ i.i.d. from same distribution
 $\Rightarrow$ Sample mean estimates $\mathbb{E}[X]$ using **multiple realizations**
at **one "location"** (trial number)

In spatial statistics:

Observe $Z(s_1), Z(s_2), \dots, Z(s_n)$
Only **one realization** at **many locations**
 $\Rightarrow$ Need **stationarity**: replace ensemble averages with spatial averages

Definition of Strict Stationarity

Strict Stationarity (Very Strong)

A spatial process is **strictly stationary** if for any set of locations $\mathbf{s}_1, \dots, \mathbf{s}_k$ and any spatial shift $\mathbf{h}$:

$$F(Z(\mathbf{s}_1), \dots, Z(\mathbf{s}_k)) = F(Z(\mathbf{s}_1 + \mathbf{h}), \dots, Z(\mathbf{s}_k + \mathbf{h}))$$

where F is the joint cumulative distribution function.

- All finite-dimensional distributions are shift-invariant
- Implies constant mean, constant variance, covariance depends only on lag
- Too strong for most real data

Second-Order (Weak) Stationarity

Definition

A process is **second-order stationary** (weakly stationary) if:

- 1 **Constant mean:** $\mathbb{E}[Z(\mathbf{s})] = \mu$ for all $\mathbf{s}$
- 2 **Constant variance:** $\text{Var}(Z(\mathbf{s})) = \sigma^2 < \infty$ for all $\mathbf{s}$
- 3 **Covariance depends only on lag:**

$$\text{Cov}(Z(\mathbf{s}), Z(\mathbf{s} + \mathbf{h})) = C(\mathbf{h}) \quad \text{for all } \mathbf{s}$$

Consequences:

- Correlation function: $\rho(\mathbf{h}) = C(\mathbf{h})/C(0)$
- $\rho(\mathbf{0}) = 1$, $\rho(\mathbf{h}) \rightarrow 0$ as $\|\mathbf{h}\| \rightarrow \infty$ (typically)
- Covariance is positive definite

What Does "Covariance is Positive Definite" Mean?

Definition (simplified)

A function $C(\mathbf{h})$ is **positive definite** if for any set of locations $\mathbf{s}_1, \dots, \mathbf{s}_n$ and any real numbers $a_1, \dots, a_n$:

$$\sum_{i=1}^n \sum_{j=1}^n a_i a_j C(\mathbf{s}_i - \mathbf{s}_j) \geq 0$$

In plain English:

- The covariance matrix $\mathbf{C}$ (where $C_{ij} = C(\mathbf{s}_i - \mathbf{s}_j)$) must have **non-negative eigenvalues**
- No linear combination of values can have negative variance
- This is not optional it's required by the laws of probability

If your covariance function is not positive definite, it's not a valid covariance function.

Positive Definite vs Positive Semi-Definite

Property	Formula	Meaning
Positive definite	$a^T \mathbf{C} a > 0$ for all $a \neq 0$	All eigenvalues positive
Positive semi-definite	$a^T \mathbf{C} a \geq 0$ for all $a \neq 0$	Some eigenvalues zero

In spatial statistics:

- **Valid** covariance functions are **positive semi-definite** (strictly speaking)
- The nugget term τ^2 makes it positive definite (if $\tau^2 > 0$)
- Linear combinations of valid functions remain valid

Example of a NON-positive definite function (invalid):

$$C(h) = \cos(h) \quad \text{for } h \in \mathbb{R}$$

It oscillates and becomes negative for some h , causing negative variances.

Simple Counterexample: An Invalid Covariance

Suppose someone proposes: $C(h) = 1 - h^2$ for $h \in [0, \infty)$

Check positivity: Take 3 equally spaced points at $s_1 = 0, s_2 = 1, s_3 = 2$

$$\mathbf{C} = \begin{bmatrix} 1 & 1 - 1^2 = 0 & 1 - 4 = -3 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$$

Choose $a = (1, 0, 1)^T$:

$$a^T \mathbf{C} a = [1, 0, 1] \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = 1 + 0 + 1 + 2(-3) = -4$$

Negative! This covariance is invalid — it can't represent any real spatial process.

Example: Is This Process Second-Order Stationary?

Consider: $Z(s) = 10 + \varepsilon(s)$ where $\varepsilon(s)$ are independent with mean 0, variance 4.

Check:

- $\mathbb{E}[Z(s)] = 10$ — constant
- $\text{Var}(Z(s)) = 4$ — constant
- For $h \neq 0$: $C(h) = \text{Cov}(Z(s), Z(s+h)) = 0$ — depends only on h

This is second-order stationary (pure nugget effect).

Practically: The mean does not change across space, and pairs at the same distance have the same covariance regardless of where they are located.

Example: Non-Stationary Process

Consider: $Z(s) = 5 + 0.1 \cdot s + \varepsilon(s)$ (linear trend in s)

Check:

- $\mathbb{E}[Z(s)] = 5 + 0.1s$ — **NOT constant** (depends on s)
- Therefore, **not second-order stationary**

Solution: Detrend the data first (e.g., regression, GAM).

The Intrinsic Hypothesis: A Weaker Alternative

Recall: Second-order stationarity can fail even when differences are well-behaved.

Intrinsic Hypothesis

A process is **intrinsic** if:

- 1 $\mathbb{E}[Z(\mathbf{s} + \mathbf{h}) - Z(\mathbf{s})] = 0$ for all $\mathbf{s}, \mathbf{h}$
- 2 $\text{Var}[Z(\mathbf{s} + \mathbf{h}) - Z(\mathbf{s})] = 2\gamma(\mathbf{h})$ depends only on $\mathbf{h}$

Key implications:

- **No assumption on mean** — only that increments have mean zero
- **No assumption on covariance** — only that variance of increments exists and is stationary
- **Variogram exists** even if covariance does not

Intrinsic vs Second-Order Stationary: An Example

The Random Walk (Brownian motion) in 1D

- Mean: $\mathbb{E}[Z(s)] = 0$ for all s (constant, so OK for second-order?)
- Variance: $\text{Var}(Z(s)) = \sigma^2|s|$ — **NOT constant** (depends on s)
- Covariance: $C(s, t) = \sigma^2 \min(s, t)$ — depends on s, t individually, not just lag

Therefore: **Not second-order stationary.**

But it IS intrinsic:

$$\mathbb{E}[Z(s+h) - Z(s)] = 0, \quad \text{Var}(Z(s+h) - Z(s)) = \sigma^2|h|$$

So the variogram exists: $\gamma(h) = \frac{\sigma^2}{2}|h|$.

The variogram is valid for intrinsic processes. Covariance is not required.

Summary Table: Stationarity Types for Spatial Data

Property	Strict	Second-order	Intrinsic
Constant mean	✓	✓	Not required (increments mean 0)
Constant variance	✓	✓	Not required
Covariance depends only on lag	✓	✓	Not required
Variance of increments depends only on lag	✓	✓	✓
Finite covariance	✓	✓	Not required
Typical in practice	Rare	Common	Very common

Most geostatistical analyses assume at least intrinsic stationarity.

What If My Data Are Not Stationary?

Common solutions:

- 1 **Detrending:** Fit a model for the large-scale variation and work with residuals

$$Z(\mathbf{s}) = m(\mathbf{s}) + \varepsilon(\mathbf{s})$$

where $m(\mathbf{s})$ is polynomial, spline, or regression on covariates

- 2 **Universal kriging:** Model the mean as a linear combination of known basis functions
- 3 **Transformation:** Box-Cox, log transform to stabilize variance (if variance changes with mean)
- 4 **Use higher-order models:** Locally stationary, spatially varying parameters

Always check stationarity before computing a variogram!

How to Check Stationarity in Practice?

Visual checks:

- Spatial map of $Z(\mathbf{s})$ — look for large-scale trends
- Scatterplot of Z vs x coordinate and vs y coordinate
- Spatial map of local means and variances (moving window)

Quantitative checks:

- Regression of Z on coordinates — if slope significant, spatial trend exists
- Compute variograms for different sub-regions — if they differ strongly, non-stationarity
- Mann-Kendall trend test for spatial data

Take-home: Stationarity is an assumption, not a fact.
Always verify (or at least acknowledge it).

Stationarity Assumptions

Second-Order (Weak) Stationarity

For all $\mathbf{s}, \mathbf{h}$:

$$\mathbb{E}[Z(\mathbf{s})] = \mu \quad (\text{constant mean})$$

$$\text{Cov}(Z(\mathbf{s}), Z(\mathbf{s} + \mathbf{h})) = C(\mathbf{h}) \quad (\text{depends only on lag})$$

Intrinsic Hypothesis (Weaker)

- $\mathbb{E}[Z(\mathbf{s} + \mathbf{h}) - Z(\mathbf{s})] = 0$
- $\text{Var}[Z(\mathbf{s} + \mathbf{h}) - Z(\mathbf{s})] = 2\gamma(\mathbf{h})$ (depends only on $\mathbf{h}$)

The variogram requires only the intrinsic hypothesis.

Definition of the Variogram

Definition

The **semivariogram** (or simply variogram) is defined as:

$$\gamma(\mathbf{h}) = \frac{1}{2} \text{Var}(Z(\mathbf{s} + \mathbf{h}) - Z(\mathbf{s}))$$

The quantity $2\gamma(\mathbf{h})$ is the **variogram**.

Properties:

$$\gamma(0) = 0 \quad (\text{by definition})$$

$$\gamma(\mathbf{h}) \geq 0$$

$$\gamma(\mathbf{h}) = \gamma(-\mathbf{h}) \quad (\text{even function})$$

$$\gamma(\mathbf{h}) = \gamma(\|\mathbf{h}\|) \quad (\text{isotropic case - distance only})$$

Relation to Covariance (Under Second-Order Stationarity)

If second-order stationarity holds, then:

$$C(\mathbf{h}) = \text{Cov}(Z(\mathbf{s}), Z(\mathbf{s} + \mathbf{h}))$$

We have:

$$\gamma(\mathbf{h}) = C(0) - C(\mathbf{h})$$

where $C(0) = \text{Var}(Z(\mathbf{s}))$ is the total variance.

Important: The variogram can exist even when the covariance does not (e.g., intrinsic processes).

Matheron's Method of Moments Estimator

Given data at locations $\mathbf{s}_1, \dots, \mathbf{s}_n$:

$$\hat{\gamma}(\mathbf{h}) = \frac{1}{2N(\mathbf{h})} \sum_{i,j:\mathbf{s}_i-\mathbf{s}_j \approx \mathbf{h}} (Z(\mathbf{s}_i) - Z(\mathbf{s}_j))^2$$

where:

- $N(\mathbf{h})$ = number of point pairs at lag $\mathbf{h}$
- The sum is over all pairs whose separation vector is approximately $\mathbf{h}$
- $\approx \mathbf{h}$ means within a chosen tolerance (lag bin)

Lag Binning:

- Define lag spacing d
- Tolerance $\pm d/2$
- Example: Lag 1: $0.5d$ to $1.5d$

Key Rules:

- 1 Minimum number of pairs per bin (e.g., $N(h) \geq 30$)
- 2 Maximum lag distance $\leq$ half the domain diameter
- 3 Irregular sampling affects bin counts

Directional vs Omnidirectional:

- Omnidirectional: all directions
- Directional: specific angle $\pm$ tolerance

Example: Empirical Variogram Calculation

Suppose 3 points on a line at positions 0, 2, 5 with values: $Z = 10, 12, 15$. Find the empirical variogram.

Why Model the Variogram?

- Empirical variogram is discontinuous (can't use directly)
- Need a valid **positive definite** function for kriging
- Models provide interpretation (range, sill, nugget)
- Enable prediction at unobserved locations

Three Key Parameters

- Nugget (C_0):** Discontinuity at origin (measurement error + micro-scale variability)
- Sill ($C_0 + C$):** Total variance where variogram flattens
- Range (a):** Distance beyond which spatial correlation disappears

Common Variogram Models - Spherical

Most widely used model, linear near origin, reaches sill exactly at range a .

$$\gamma(h) = \begin{cases} C_0 + C \left[1.5 \frac{h}{a} - 0.5 \left(\frac{h}{a} \right)^3 \right] & \text{for } 0 < h \leq a \\ C_0 + C & \text{for } h > a \end{cases}$$

Shape: Parabolic at origin (smooth), then linear, then flat

Common Variogram Models - Exponential & Gaussian

Exponential:

$$\gamma(h) = C_0 + C \left[1 - \exp \left(-\frac{h}{a} \right) \right]$$

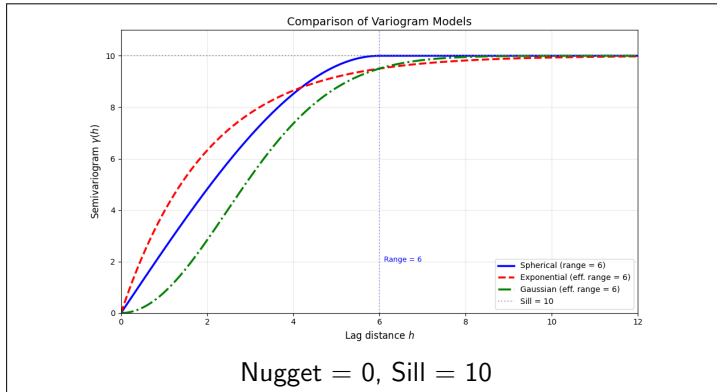
- Approaches sill asymptotically
- **Effective range:** $\approx 3a$ (where 95% of sill is reached)
- Linear near origin

Gaussian:

$$\gamma(h) = C_0 + C \left[1 - \exp \left(-\frac{h^2}{a^2} \right) \right]$$

- Parabolic near origin (very smooth)
- Effective range: $\approx \sqrt{3}a$
- **Caution:** Can cause numerical instability

Visual Comparison of Models



Model	Origin behavior	Sill	Whether range exact
Spherical	Linear	Exact	Yes
Exponential	Linear	Asymptotic	No (effective)
Gaussian	Parabolic	Asymptotic	No

Adding Nugget to Any Model

The nugget C_0 represents:

- Measurement error
- Spatial variability at distances smaller than the minimum sampling distance
- Micro-scale structure

$$\gamma(h) = C_0 + (C_1 - C_0) \cdot \text{model}(h)$$

where C_1 is the total sill.

Important: A pure nugget model $\gamma(h) = C_0$ for $h > 0$ means no spatial correlation (white noise).

Geometric vs Zonal Anisotropy

Geometric Anisotropy

Range varies with direction, sill is constant.

$$\gamma(\mathbf{h}) = \gamma\left(\sqrt{\mathbf{h}^T \mathbf{A} \mathbf{h}}\right)$$

where $\mathbf{A}$ is an anisotropy matrix (elliptical stretching).

Zonal Anisotropy

Sill varies with direction (less common, more complex).

Common fix: Transform coordinates $\mathbf{s}' = \mathbf{A}^{-1/2}\mathbf{s}$ to restore isotropy.

Top 7 Mistakes in Variography

- 1 Computing variogram without checking stationarity (trends cause inflation)
- 2 Ignoring the proportional effect (variance increases with mean)
- 3 Over-interpreting noise at small lags (limited pairs)
- 4 Choosing a model with range larger than maximum sampling distance
- 5 Using the variogram for prediction without cross-validation
- 6 Assuming isotropy without checking
- 7 Forgetting that the variogram is not a correlation (values can exceed total variance)

Checking Your Variogram Model - Cross-Validation

Leave-one-out cross-validation

For each observation i :

- Predict $\hat{Z}(\mathbf{s}_i)$ using all other data points
- Compute standardized residual:

$$r_i = \frac{Z(\mathbf{s}_i) - \hat{Z}(\mathbf{s}_i)}{\sigma_i}$$

where σ_i is the kriging standard error

Desirable properties:

- Mean residual close to 0
- Variance of residuals close to 1
- No trends in residuals vs predicted values

Summary Table for Reference

Term	Meaning
$\gamma(h)$	Semivariogram at lag h
Nugget (C_0)	Discontinuity at origin (noise + micro-scale)
Partial Sill (C)	Structural variance
Sill ($C_0 + C$)	Total variance (plateau)
Range (a)	Distance beyond which no correlation
Intrinsic hypothesis	Variance of increments is stationary
$2N(h)$	Twice the number of pairs (denominator)

Homework (Optional)

Practical Exercise

Given a dataset of 50 points with coordinates (x,y) and measured zinc concentration (simulated or real):

- 1 Plot the data and check for trends
- 2 Compute the omnidirectional empirical variogram with 6-8 lags
- 3 Fit a spherical model manually (using least squares or visual fitting)
- 4 Report: nugget, sill, range
- 5 Compute directional variograms ($0^\circ, 45^\circ, 90^\circ$) is there anisotropy?

Deliverable: 1-page report + variogram plot

-  Cressie, N. (2015). *Statistics for Spatial Data*. Wiley.
-  Webster, R., & Oliver, M.A. (2007). *Geostatistics for Environmental Scientists*. Wiley.
-  Matheron, G. (1963). Principles of geostatistics. *Economic Geology*, 58(8), 1246-1266.